

American Joint Committee on Cancer Classification of Posterior Uveal Melanoma (Tumor Size Category) Predicts Prognosis in 7731 Patients

Carol L. Shields, MD,¹ Swathi Kaliki, MD,¹ Minoru Furuta, MD,¹ Enzo Fulco, MD,¹ Carolina Alarcon, MD,¹ Jerry A. Shields, MD¹

Purpose: To evaluate the clinical features and prognosis of posterior uveal (ciliary body and choroid) melanoma based on the American Joint Committee on Cancer (AJCC) classification (7th edition) of primary tumor (T).

Design: Retrospective, interventional case series.

Participants: Seven thousand seven hundred thirty-one patients.

Intervention: Ocular management including plaque radiotherapy, enucleation, local resection, or laser therapy.

Main Outcome Measures: Melanoma-related metastasis and death.

Results: Of 7731 patients with posterior uveal melanoma, the AJCC classification based on T was category T1 in 3557 (46%), T2 in 2082 (27%), T3 in 1599 (21%), and T4 in 493 (6%). Based on tumor categories T1, T2, T3, and T4, respectively, features that showed significant increase with tumor category included patient age at presentation (57, 58, 58, and 61 years; $P < 0.001$), tumor base (8, 12, 15, and 20 mm; $P < 0.001$), tumor thickness (3.5, 5.2, 8.9, and 11.4 mm; $P < 0.001$), mushroom configuration (8%, 20%, 38%, and 39%; $P < 0.001$), associated subretinal fluid (64%, 80%, 82%, and 83%; $P < 0.001$), intraocular hemorrhage (5%, 12%, 17%, and 18%; $P < 0.001$), rupture of Bruch's membrane (9%, 24%, 40%, and 40%; $P < 0.001$), and extraocular extension (1%, <1%, 4%, and 12%; $P < 0.001$). After therapy, Kaplan-Meier estimates of metastasis at 5, 10, and 20 years were 8%, 15%, and 25% for category T1, 14%, 25%, and 40% for category T2, 31%, 49%, and 62% for category T3, and 51%, 63%, and 69% for category T4, respectively ($P < 0.001$). Kaplan-Meier estimates of death at 5, 10, and 20 years were 4%, 8%, and 11% for category T1, 8%, 13%, and 24% for category T2, 19%, 27%, and 36% for category T3, and 30%, 43%, and 51% for category T4, respectively ($P < 0.001$). Compared with category T1, the hazard ratio for metastasis and death for T2 was 1.8 and 1.9, respectively, that for T3 was 4.5 and 4.7, respectively, and that for T4 was 8.2 and 8.8, respectively.

Conclusions: Based on the AJCC classification, increasing tumor category was associated with older age, larger tumor, and greater incidence of subretinal fluid, hemorrhage, and extraocular extension. Compared with uveal melanoma classified as T1, the rate of metastasis and death was 2 times greater for T2, 4 times greater for T3, and 8 times greater for T4. The risk for metastasis and death increased 2-fold with each increasing melanoma category.

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Posterior uveal melanoma (ciliary body [CB] and choroidal melanoma) represents 96% of all uveal melanoma.^{1,2} Management of melanoma is focused on early detection,^{3,4} with therapeutic interventions of enucleation, radiotherapy, resection, and forms of laser photocoagulation, transpupillary thermotherapy, and photodynamic therapy.² Despite local therapeutic efforts, systemic prognosis remains guarded because melanoma can metastasize early in the course of disease⁵ and primarily is dependent on tumor genetics and size.⁶⁻¹¹

Posterior uveal melanoma generally is classified into size categories based on thickness, including small (≤ 3 mm), medium ($>3-8$ mm), and large (>8 mm), with prognosis directly related to tumor size.^{11,12} Estimates of 10-year mortality revealed 12% for small melanoma, 26% for medium melanoma, and 49% for large melanoma.¹¹ More precise risk estimates based on ultrasonographically measured tumor thickness in millimeters disclosed that each 1-mm gain imparted approximately a 5% increased risk for metastasis.¹¹